

BRISTOL WEATHER CENTRE, United Kingdom

WMO: 037260

Lat: 51.467N

Lon: 2.600W

Elev: 11

StdP: 101.19

Time Zone: 0.00 (EUW)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.9	(c) -0.5	(d) -6.6	(e) 2.2	(f) 1.6	(g) -4.8	(h) 2.5	(i) 2.4	(j) 12.8	(k) 9.0	(l) 11.6	(m) 9.5	(n) 2.1	(o) 70	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.9	(c) 26.7	(d) 18.3	(e) 24.8	(f) 17.2	(g) 23.0	(h) 16.6	(i) 19.3	(j) 24.3	(k) 18.2	(l) 22.6	(m) 17.4	(n) 21.2	(o) 3.2	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.5	20.8	16.6	11.8	19.9	15.8	11.2	19.0	54.8	24.5	51.5	22.7	48.7	21.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.8	7.7	-4.0	29.4	1.3	2.5	-4.9	31.2	-5.7	32.7	-6.4	34.1	-7.3	36.0
			-4.8	20.8	1.6	0.9	-6.0	21.4	-7.0	21.9	-7.9	22.4	-9.1	23.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	6.1	6.6	8.2	9.4	13.1	15.4	17.8	18.0	15.1	11.7	8.7	6.2
	DBStd	5.05	3.02	3.00	2.37	2.49	3.02	2.47	2.32	2.54	2.06	2.66	2.84	3.48
	HDD10.0	526	122	99	63	40	6	0	0	0	0	14	57	124
	HDD18.3	2594	378	329	313	269	163	94	39	37	100	206	289	377
	CDD10.0	1032	2	3	8	21	103	163	241	248	152	66	18	6
	CDD18.3	59	0	0	0	0	2	7	22	27	2	0	0	0
	CDH23.3	338	0	0	0	0	15	49	117	149	8	0	0	0
	CDH26.7	59	0	0	0	0	0	8	13	38	0	0	0	0
Wind	WSAvg	3.3	3.8	3.7	3.6	3.5	3.2	3.2	3.1	2.7	2.7	3.1	3.0	3.6
Precipitation	PrecAvg	894	91	62	69	58	61	66	60	77	85	88	91	95
	PrecMax	1122	175	145	162	125	150	158	150	181	200	223	210	212
	PrecMin	634	10	2	4	3	8	12	8	7	21	4	32	20
	PrecStd	120	47	39	34	30	35	41	35	44	46	51	42	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.4	16.7	19.9	25.1	27.8	28.2	30.6	24.7	19.7	15.7	14.0
		MCWB	10.8	11.0	11.4	12.2	15.7	19.0	19.3	18.9	18.2	15.8	13.8	12.3
	2%	DB	12.1	12.5	14.0	16.7	23.2	24.5	26.5	27.2	21.8	17.5	14.6	13.1
		MCWB	10.2	10.2	10.1	11.1	15.6	17.1	18.3	18.4	16.4	14.4	13.0	11.7
	5%	DB	11.2	11.2	12.6	14.7	21.0	22.1	24.5	24.4	19.9	16.5	13.7	12.1
		MCWB	9.7	9.2	9.5	10.4	14.7	15.6	17.1	17.5	15.2	13.9	12.1	10.7
	10%	DB	10.4	10.5	11.6	13.2	18.5	20.0	22.5	22.4	18.6	15.6	12.9	11.1
		MCWB	8.9	8.7	9.0	9.8	13.5	14.6	16.7	16.6	14.6	13.4	11.4	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.7	12.2	14.0	17.7	19.9	24.0	20.2	18.6	16.3	14.2	12.7
		MCDB	12.6	13.7	15.1	17.6	22.8	26.3	24.0	26.5	23.1	18.6	15.2	13.5
	2%	WB	10.7	10.6	11.2	12.0	16.4	17.4	19.2	19.3	17.3	15.3	13.3	11.8
		MCDB	11.8	12.0	13.0	14.6	21.3	23.1	23.9	24.5	20.3	17.0	14.3	12.8
	5%	WB	9.9	9.7	10.4	11.1	15.1	16.2	18.0	18.5	16.5	14.5	12.5	11.0
		MCDB	11.0	11.0	11.9	13.9	19.7	20.7	21.8	22.8	18.9	16.1	13.5	11.9
	10%	WB	9.1	8.9	9.7	10.3	14.0	15.2	17.1	17.5	15.6	13.7	11.5	10.0
		MCDB	10.3	10.3	11.0	12.6	18.2	18.9	21.1	21.2	17.8	15.4	12.7	11.1
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.7	5.3	6.2	7.3	6.9	7.0	6.9	6.1	5.4	4.5	4.1
		MCDBR	4.8	5.2	6.7	8.4	11.2	10.9	11.0	10.4	8.1	5.9	4.3	4.4
		MCWBR	4.5	4.3	4.2	4.6	5.5	5.4	5.3	4.0	4.3	4.0	3.8	4.4
	5% WB	MCDBR	4.7	5.0	5.3	7.1	9.8	9.4	8.7	8.4	7.0	5.3	4.3	4.5
		MCWBR	4.7	4.7	4.1	4.5	5.3	5.3	5.1	4.1	4.8	4.3	4.1	4.8
Clear-Sky Solar Irradiance	taub	0.330	0.344	0.378	0.399	0.403	0.396	0.393	0.380	0.370	0.356	0.330	0.319	
	taud	2.349	2.370	2.276	2.236	2.262	2.315	2.330	2.383	2.405	2.416	2.399	2.329	
	Ebn at Noon	672	774	810	837	851	859	855	849	815	750	670	617	
	Edh at Noon	64	82	107	125	128	122	119	107	94	77	60	56	
All-Sky Solar Radiation	RadAvg	0.78	1.48	2.62	4.03	4.95	5.37	5.03	4.17	3.14	1.78	0.94	0.61	
	RadStd	0.07	0.18	0.33	0.46	0.25	0.57	0.47	0.37	0.23	0.16	0.09	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page